

PRESS RELEASE

**Entrepreneurship Development and Innovation Institute-TN
“5-Days Laptop Hardware & Chip-Level Training Programme”**

Entrepreneurship Development and Innovation Institute (EDII-TN), Chennai is organizing a 5-day Entrepreneurship Development programme on **“Laptop Hardware & Chip-Level Training Programme”** from **15.06.2026 to 19.06.2026 (10.00 A.M. to 05.00 P.M.)** at EDII-TN Campus, Guindy Chennai-600032.

Programme Highlights:

Day 1:

- Laptop Hardware Fundamentals & System Installation
- Introduction to Laptop Architecture and Components
- Laptop Assembly and Disassembly Techniques
- Laptop Adapter Testing and Troubleshooting
- RAM and ROM: Types, Functions, and Applications
- CPU Sockets and Processor Technologies
- Motherboard Types, Chipsets, and Architecture
- Hardware Troubleshooting Techniques
- Understanding Error Codes and Diagnostic Methods
- Operating System Installation and Driver Configuration

Day 2:

- Chip-Level Components & Soldering Techniques
- Introduction to Chip-Level Servicing
- SMT (Surface Mount Technology) Components: Identification and Applications
- Advanced Chip-Level Components Training and Fault Diagnosis
- Study of Electronic Components:
 - Resistors
 - Capacitors
 - Inductors
 - Diodes
 - NPN & PNP Transistors
 - UJT, FET, and MOSFET
- Component Construction, Working Principles, and Practical Demonstrations
- Component Removal and Replacement Techniques (Hands-on Practice)
- Fault Identification Using Digital Multimeter
- Individual Practical Assessment

Day 3:

- Power Management & Laptop Voltage Analysis
- Fundamentals of Voltage, Current, Amperes, and Wattage
- Working Principles of NPN, PNP, FET, UJT, and MOSFET
- N-Channel and P-Channel MOSFET Analysis and Fault Detection
- Laptop Power Rail Analysis:
 - USB Ports
 - CPU Fan
 - Power IC
 - HDD/SSD
 - DVD Drive
 - BIOS IC

- CMOS Battery
 - Keyboard
 - Power Switch
 - Wi-Fi Module
 - Processor
 - RAM
 - Speakers
 - Chipset
 - LED Indicators
 - Touchpad
 - Camera
 - Wi-Fi Switch
 - Voltage Measurement and Troubleshooting Techniques
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Day 4:

- Motherboard Diagnostics & Schematic Analysis
 - Understanding Motherboard Circuit Diagrams
 - Voltage Tracing Using Schematics
 - Motherboard Circuit Tracking and Analysis
 - Diagnostic Procedures for Dead Laptops
 - Advanced Troubleshooting Methodologies
 - Practical Fault-Finding Exercises
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Day 5:

- Signal Tracing & BIOS Programming
- Charging Circuit Analysis and Troubleshooting
- Standby Power Section Analysis
- Signal Tracing and Diagnostics:
 - SUS_CLK
 - RTC_RESET
 - INTRUDER
 - INTVREN
 - AC_OK
 - S5_ENABLE
 - PGOOD
 - RSMRST
 - PWRBTNOUT
 - SLP_S4
 - SLP_S3
- BIOS Flashing and Recovery Procedures
- Practical Troubleshooting and Assessment
- Course Review and Certification Guidance

Training Methodology

- Classroom Instruction
- Live Demonstrations
- Hands-on Practical Sessions
- Individual Performance Evaluation
- Real-Time Fault Diagnosis Exercises

Learning Outcomes

Upon successful completion of this programme, participants will be able to:

- Assemble and disassemble laptops safely.
- Diagnose and repair hardware faults.
- Perform chip-level troubleshooting and component replacement.
- Trace motherboard circuits using schematics.
- Analyze power and signal sections of laptop motherboards.
- Flash and recover BIOS firmware.

- Independently troubleshoot and repair laptop hardware issues.

Eligibility: Candidates (Male/Female/Transgender) aged 18 years above with a minimum qualification of 10th Standard are eligible to apply.

Government Certificate will be provided upon completion.

Hostel facility is available at affordable rates (first-come, first-served basis). Pre-registration is mandatory for more information, please visit: www.editn.in Address Entrepreneurship Development and Innovation Institute. No 1, EDI Institute Road, SIDCO Industrial Estate, Ekkaduthangal, Guindy, Chennai 600032.

For assistance - Contact us on working days (Monday to Friday), between 10:00 AM and 5:45 PM at:  **8668100181 / 9360221280**

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